

Tech Leaders · Class 15

# Parametric Robot Gripper Design

Design once, resize forever

Parameters

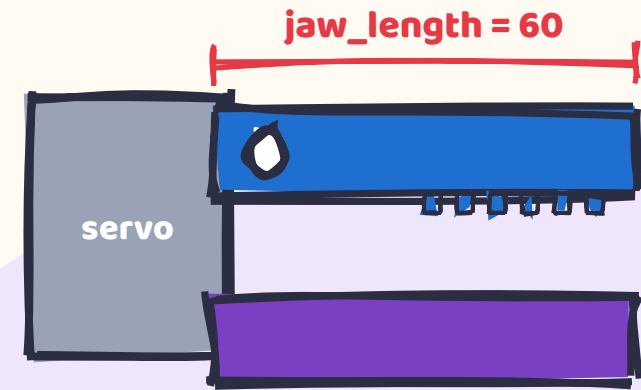
Constraints

Onshape / OpenSCAD

Print & test

Did you know?

Car makers design one parametric seat frame, then change a few numbers to fit dozens of different car models.





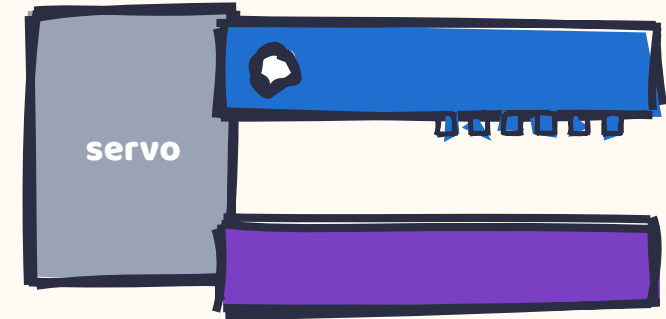
Let's think...

# One gripper for an egg AND a bottle?



## Different jobs

A factory robot must pick up eggs today and juice bottles tomorrow. Redrawing the gripper from scratch each time takes days. What if the design were built from **numbers**, so changing 'object\_width = 45' to '70' redesigned the whole thing in seconds?



Buy a new robot?

Redraw it?

Change a number!

**Today:** you design a **parametric gripper finger**, make two variants for different objects, print and test them.





## Learning objectives

# By the end of this class I can...



1

**Parametric idea:** explain **parameters**, **constraints** and **design intent****Proof:** show which numbers drive our model

2

**Requirements:** turn the object to grip into **measured requirements** and **parameters****Proof:** a parameter table with units and limits

3

**Model:** build a **fully constrained** gripper finger driven by variables in Onshape (or OpenSCAD)**Proof:** changing one variable resizes the whole part correctly

4

**Variants:** generate **two variants** for different objects by changing parameters only**Proof:** two STL files from one model

5

**Test & iterate:** print, test the grip and **improve** one parameter**Proof:** a test log and our version 2 value



Objective 1 • Parametric idea • video

# Video time!



## What is Parametric & Non-Parametric Design?

Boldform Consulting • about 5 minutes • [youtu.be/ERuOe8eLHOE](https://youtu.be/ERuOe8eLHOE)

### Watch for...

- 1 What is the difference between **parametric** and **direct** (non-parametric) modelling?
- 2 What is a **constraint**?
- 3 When might non-parametric modelling be better?



**After the video:** Name an object in the room that comes in many sizes. Which 2 or 3 numbers would you make parameters?





## Objective 1 • Parametric idea

# Parameters + constraints = design intent



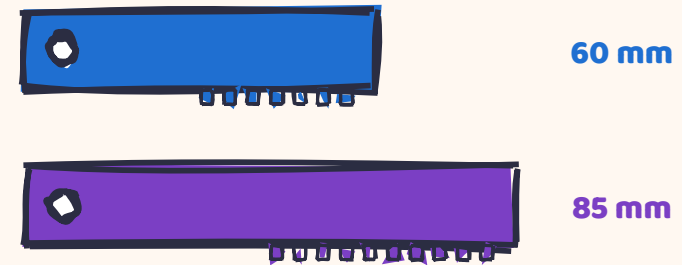
## The idea

**Parameter:** a named number that drives the model (jaw\_length = 60).

**Constraint:** a rule that must stay true (hole is **centred**, edges are **parallel**, ridge spacing is **equal**).

**Design intent:** what should happen when a number changes. Make the finger longer: the hole stays near the pivot, the ridges spread along the tip.

## Example: a recipe that scales



A cake recipe for 4 people uses 2 eggs and 200 g flour. For 8 people you double everything. The ratios (constraints) stay the same; one number (people) drives the rest. That is a parametric recipe!

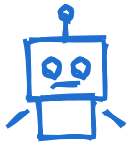




Real world



# Parametric design in industry



## Robot tooling

Grippers resized for each new product on the line



## Cars

One part family, many models



## Architecture

Stadium roofs and towers generated from parameters



## Medical

Prosthetic hands sized from a patient's measurements



e-NABLE volunteers 3D print free prosthetic hands. Why is parametric design perfect for that?

## Customisation

Measure the person, type the numbers, print a hand that fits.



## Class game

# Human CAD



### How to play

1. Team A writes instructions to draw a gripper finger on squared paper, using ONLY parameters: L, W, hole\_d.
2. Example: 'rectangle L by W; hole of diameter hole\_d, centred, 8 from the left end'.
3. Team B draws it with L=10, W=3. Then again with L=16, W=4.
4. Does the second drawing still look right?

### Big idea

If the design breaks when a number changes, a constraint is missing.



### Then discuss

Where did the instructions break when the numbers changed? What constraint was missing?





## Objective 2 • Requirements • plan it

# From object to parameters



| Parameter    | Value | Unit | Why / limit                    |
|--------------|-------|------|--------------------------------|
| object_width | 45    | mm   | width of the egg or bottle     |
| jaw_length   | 60    | mm   | reach past the object's middle |
| jaw_width    | 12    | mm   | $\geq 10$ for strength in PLA  |
| thickness    | 5     | mm   | matches servo horn             |
| pivot_d      | 3.2   | mm   | M3 screw + 0.2 clearance       |
| ridge_depth  | 1.5   | mm   | grip without crushing          |

## Measure first

Use calipers on the real object and on the **SG90 servo** (body about  $22.5 \times 12.2$  mm). Never trust an online drawing alone.



## Units!

Write mm on everything. A gripper 45 cm wide is a famous classroom mistake.



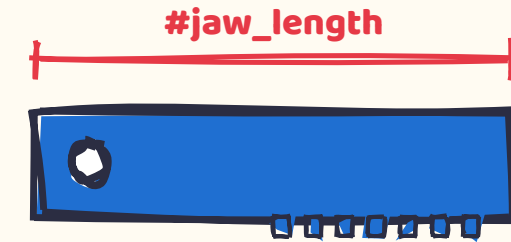


## Objective 3 · Build step 1 of 3

# Onshape: variables and a sketch



- a** Sign in to **Onshape** (free education account via the teacher). New document → Part Studio.
- b** Add a **Variable** feature for each parameter: **#jaw\_length** = 60 mm, **#jaw\_width** = 12 mm ...
- c** Sketch on the Top plane: a rectangle and a circle for the pivot.
- d** Dimension with variables: type **#jaw\_length** instead of 60.
- e** Add constraints until the sketch turns **black** (fully defined).



**Check:** Every line is black and every size is a variable.



Blue lines = not constrained. They will move unexpectedly when you change a number.



## Objective 3 · Build step 2 of 3

# Extrude, pattern, fillet



- a** Extrude the sketch by **#thickness**.
- b** Sketch one ridge on the gripping edge, extrude it by **#ridge\_depth**.
- c** **Linear pattern** the ridge: count = **floor(#jaw\_length / 8)**, so longer jaws get more ridges.
- d** **Fillet** outer corners 1 mm: stronger and kinder to fingers.
- e** Right-click the part → **Export** → STL.



**Check:** The finger has evenly spaced ridges along the tip and a clean pivot hole.





## Objective 3 • Model • code version

# The same finger as code (OpenSCAD)



```
// Parametric gripper finger (OpenSCAD)
jaw_length  = 60;      // mm
jaw_width   = 12;
thickness   = 5;
ridge_depth = 1.5;
pivot_d     = 3.2;    // M3 + clearance

difference() {
  union() {
    cube([jaw_length, jaw_width, thickness]);
    for (x = [jaw_length * 0.5 : 4 : jaw_length - 4])
      translate([x, -ridge_depth, 0])
        cube([2, ridge_depth, thickness]);
  }
  translate([6, jaw_width / 2, -1])
    cylinder(d = pivot_d, h = thickness + 2, $fn = 40);
}
```

## Code = CAD

OpenSCAD (free) builds 3D models from code. The parameters are just variables at the top.

## Customizer

Window → Customizer shows the variables as sliders. Change them and watch the model update.

## Why both?

Onshape is how industry works. OpenSCAD shows that a model can be a program. Try either!





## Objective 4 • Variants



# Variants: one model, many parts

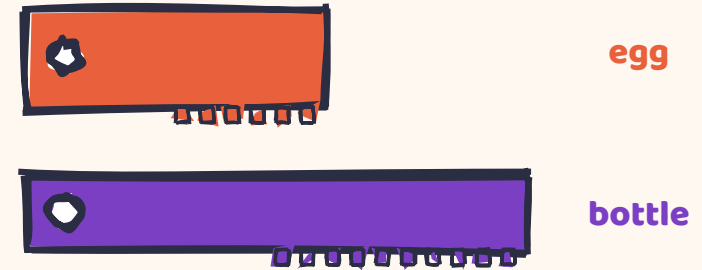
## The idea

Make two versions **without redrawing anything**:

- **Egg**: object\_width 45, ridge\_depth 0.8 (gentle), jaw\_width 14 (spread the force)
- **Bottle**: object\_width 70, jaw\_length 80, ridge\_depth 2 (more grip)

In Onshape use a **Configuration** so both live in one part. Export each as its own STL.

## Example: T-shirt sizes



A shirt pattern is drawn once, then 'graded' to S, M, L and XL by changing a few measurements. Shops don't redesign the shirt for each size.

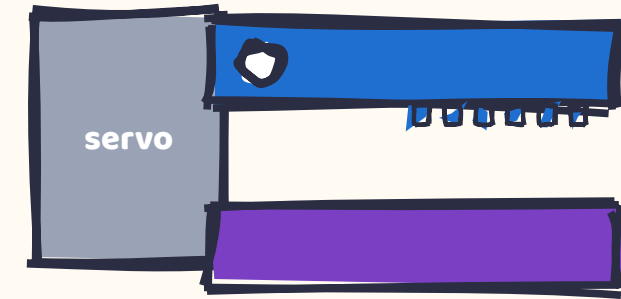


## Objective 5 · Build step 3 of 3

# Print and assemble



- a** Slice in Cura: PLA, 0.2 mm layers, **40% infill** (grippers need strength), print **flat**.
- b** Print one finger first as a test. About 20 minutes.
- c** Check the pivot hole with an M3 screw. Too tight? Change **pivot\_d**, don't drill!
- d** Fix the finger to the servo horn with the small screw.
- e** Mount the second finger on the fixed side.



**Check:** The finger fits the servo horn and the pivot screw turns freely.



Hole too tight is a parameter problem. Fix it in the model so every future print is right.



## Objective 5 · Test & iterate · test it

# Grip test



```
#include <Servo.h>
Servo grip;
int OPEN = 100, CLOSE = 40;    // tune for your gripper

void setup() { grip.attach(9); grip.write(OPEN); delay(1000); }

void loop() {
    grip.write(CLOSE); delay(2000);    // grip
    grip.write(OPEN);  delay(2000);    // release
}
```

| Object                  | Holds? | Slips? | Damage? |
|-------------------------|--------|--------|---------|
| Egg (or ping-pong ball) |        |        |         |
| Bottle                  |        |        |         |
| Pencil                  |        |        |         |
| Shake test (5 s)        |        |        |         |

### Don't stall

If the servo buzzes while holding, CLOSE is too tight: it will overheat. Open by 5°.





Objective 5 · Test & iterate · iterate

# Version 2: change ONE parameter



| Problem found   | Parameter   | Change    |
|-----------------|-------------|-----------|
| Egg slipped     | ridge_depth | 0.8 → 1.2 |
| Bottle slipped  | jaw_length  | 80 → 90   |
| Finger bent     | thickness   | 5 → 6     |
| Pivot too tight | pivot_d     | 3.2 → 3.4 |

**Engineering habit**

Change one thing, re-test, record. Then the next. If you change 3 things at once you don't know which one helped.





## Test like an engineer

# Design log



| Item                       | Result | Notes |
|----------------------------|--------|-------|
| Parameters defined (count) |        |       |
| Sketch fully constrained?  |        |       |
| Variant A / B exported     |        |       |
| Print time and grams       |        |       |
| Grip test score            |        | / 4   |
| V2 change and result       |        |       |

### Debug checklist

Model breaks when you change a number? A dimension or constraint is missing: look for blue lines.  
Pattern count error? Use floor() so it is a whole number.



Good parametric models survive extreme values. Try jaw\_length = 30 and 120. Does anything break?





Think about it

# Design for people, not just objects



## Scenario

Your parametric gripper could become a **prosthetic hand** for a child, sized from three measurements. Free files mean anyone can print one.

## Discuss

- Who checks it is strong and safe enough?
- What if a print fails while someone is using it?
- Should medical designs be open for anyone to change?

## Real example

e-NABLE volunteers worldwide have printed thousands of free hands, with careful testing and fitting guides.





Level up!

# Design challenges



## Choose your challenge



**Geared jaws:** add two spur gears (Onshape Spur Gear feature) so both jaws move together.



**Fin-ray finger:** research the fish-fin finger that bends around objects.



**Soft pads:** design slots for rubber-band pads. Better grip on smooth bottles?



**Fit the arm:** make the mount fit last term's robot arm with parameters for its holes.

## Think like an engineer

Why do soft grippers handle fruit better than hard metal ones?



## Boss level

Make ONE model with a configuration for egg, bottle and pencil, and prove all three work.





## Objective check

# Can I really do it? Prove it.



**01** What is the difference between a parameter and a constraint? Give one of each from your model.



**02** Show your parameter table. Which value came from measuring the real object?



**03** Change jaw\_length live. Does everything update correctly?



**04** Show your two variants. What changed between them?



**05** Read your test log and your version 2 change.



### Rate yourself

Colour one circle each. Red: not yet. Amber: with help. Green: I can explain it to a friend.

**Teacher:** Design review: teams swap models and try to 'break' them with extreme parameter values.





Show what you know

# What did we learn?

- ☒ Parameters are named numbers that drive a model
- ☒ Constraints keep the design intent when numbers change
- ☒ Fully constrained sketches update predictably
- ☒ Variants come from parameters, not redrawing
- ☒ Print, test, change one parameter, repeat



**Exit ticket:** your pivot hole is too tight. What do you change, and why not just drill it bigger?



## At home

Measure three cups or bottles at home. Which parameter values would your gripper need for each?



**Next class:**  
Motion-  
Controlled Game!

